

MD ENAMUL HOQ

Machine Learning Engineer — Medical Imaging AI | Foundation Models | Clinical Translation & FDA-Ready Validation

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PROFESSIONAL SUMMARY

Machine learning engineer and biomedical-imaging scientist (Ph.D., Biomedical Informatics — Imaging Track) with 5+ years building, training, validating, and deploying deep-learning and foundation-model systems for medical imaging across CT and digital pathology. Designs 2D and 3D architectures (CNNs, Vision Transformers, self-supervised foundation models) and scalable data-to-deployment pipelines over 50M+ imaging instances, translating models into clinically interpretable, regulatory-aware tools. Deep expertise in DICOM/PACS workflows, multi-institutional and federated datasets, threshold-aware evaluation, and reproducible, auditable experimentation aligned with FDA 510(k)/De Novo pathways and Good Machine Learning Practice (GMLP). Peer-reviewed author at NeurIPS, MIDL, and imaging-informatics venues; federated-learning hackathon champion; and daily user of LLM-based coding assistants (Claude, Codex) to accelerate high-quality, well-tested development.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Model development, training, validation, deployment; supervised, self-supervised & representation learning; classification, detection & quantitative image analysis; imbalanced-data learning; transfer learning & fine-tuning

Model Architectures: 2D & 3D CNNs; Vision Transformers (ViT); self-supervised foundation models (RAD-DINO, DINOv2); diffusion models; multimodal Vision-Language Models (Vision-LLMs); detection frameworks

Medical Imaging & Standards: CT; chest-radiograph (X-ray) foundation models; digital pathology (WSI); DICOM, PACS, DICOMweb; volumetric & density quantification; quantitative imaging biomarkers; RADLEX, SNOMED; 3D Slicer, ITK-SNAP; TCIA, TCGA, NCI Imaging Data Commons

Programming: Python (primary), C/C++, Java, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Keras; NVIDIA FLARE (federated learning)

Evaluation & Statistics: ROC/AUC, sensitivity/specificity, precision/recall, PR/F1; calibration; threshold & cut-point optimization; reproducibility, drift & stability analysis

MLOps & Engineering Practice: Git & collaborative development; code review; testing; experiment tracking & automated reporting; data/model/code versioning & traceability; LLM-based coding assistants (Claude, Codex)

Compute & Cloud: Linux; Slurm; HPC with A100 multi-GPU clusters; Google Cloud Platform (GCP); AWS; BigQuery

Data Engineering & Curation: Large-scale imaging pipelines; leakage-safe patient-level splitting; multi-institutional harmonization & quality control; metadata normalization; data governance (DUAs, PHI / de-identification checks)

Regulatory, Quality & Translation: FDA 510(k) & De Novo-oriented documentation; Good Machine Learning Practice (GMLP); reproducible, auditable validation; clinical-deployment & continuous-monitoring evaluation; IRB-approved prospective validation

EDUCATION

Ph.D., Biomedical Informatics (Imaging Track) — University of Arkansas for Medical Sciences (UAMS)

Expected May 2026

Little Rock, AR • Dissertation: Foundation Models for Lung Cancer Screening CT (LCS-DINO / Native-Resolution 2D)

M.S., Engineering Physics — Southeastern Louisiana University

Dec 2020

Hammond, LA • Thesis: Image Correlation for LDPE Film Deformation (computational imaging, MATLAB)

B.S., Mechatronics Engineering (Biomedical Track) — World University of Bangladesh

Apr 2014

Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (Ph.D. Researcher) — Medical Imaging AI

07/2021 – Present

University of Arkansas for Medical Sciences (UAMS) • Little Rock, AR

- Built an **end-to-end lung cancer screening pipeline** — DICOM/PACS-compliant curation → RAD-DINO embedding generation → training → patient-level evaluation — over **20M+ CT slices**, forming the core of the LCS-DINO foundation-model system.
- Demonstrated that **native-resolution 2D foundation-model embeddings** are a compute-efficient alternative to full 3D volumetric training for lung-cancer classification (first-author study, *Journal of Imaging Informatics in Medicine*, 2025).

- Designed **threshold-aware evaluation** (ROC/AUC, PR/F1, prevalence-matched optimal cut-points) reflecting real-world screening prevalence; **reduced false-positive follow-ups** in experimental cohorts.
- Scaled multi-GPU training and inference on A100 clusters with Slurm; automated experiment tracking, versioning, and report generation for reproducible, auditable results.
- Curated and governed large national cohorts in TCIA and the NCI Imaging Data Commons — cohort definition, manifest generation, DICOMweb/gsutil transfer, BigQuery access, RADLEX/SNOMED harmonization, and DUA/PHI governance.
- Collaborated with **15+ radiologists and clinicians** to translate model outputs into clinically interpretable metrics and structured reporting.
- Mentored **3 Ph.D. students and 1 M.D. student** in GPU/HPC workflows, DICOM/PACS handling, experimental design, and reproducible research.

Site Lead — Multi-Institutional Federated LDCT Screening (Site 2)

06/2025 – Present

UAMS · with Moffitt, Vanderbilt, UCLA & University of Pittsburgh · Little Rock, AR

- Lead UAMS as Site 2 in a **five-institution federated-learning program** for low-dose CT lung-cancer screening using NVIDIA FLARE.
- Architected **privacy-preserving training workflows** (secure model updates, cross-site evaluation) that improve detection performance without sharing raw PHI.
- Curated and harmonized multi-site datasets with leakage-safe splits and AI-ready pipelines to ensure cross-institution generalizability and robustness.
- Approach recognized as **Champion in 3 of 5 categories** at the CMU–NVIDIA Federated Learning Hackathon, 2026.

AI & Informatics Research Intern

02/2025 – 07/2025

Mayo Clinic · Rochester, MN

- Developed a **multimodal pathology retrieval & auto-captioning framework** using vision foundation models and LLMs — *Top-Ten Poster, Mayo Clinic AI Summit 2025*.
- Raised semantic alignment of generated captions to **>0.82 cosine similarity** via a hybrid rule-based + LLM summarization pipeline, and introduced retrieval-guided generation to reduce unsafe/hallucinated captions.
- Co-led a workshop on fine-tuning and deploying Vision-LLMs in clinical research workflows.

Graduate Research & Teaching Associate; Lab Manager

08/2019 – 12/2020

Southeastern Louisiana University · Hammond, LA

- Developed MATLAB pipelines for digital image correlation (computational imaging) with visualization and modeling tools.
- Supported 13+ researchers with debugging/optimization; instructed Physics 101/102 laboratory courses.

Research Intern — Multimedia Lab

09/2018 – 02/2019

National Chung Cheng University · Chiayi, Taiwan

- Developed depth-estimation algorithms for 2D→3D video conversion, improving perceived depth and visual quality.

R&D Engineer

08/2017 – 07/2018

TN Solution · Karlsruhe, Germany & Ho Chi Minh City, Vietnam

- Led image/signal-processing and IoT projects; improved PCB fault diagnosis through noise reduction and feature-extraction techniques.

FLAGSHIP PROJECTS & FRAMEWORKS — WHAT, HOW & MEASURED IMPACT

LCS-DINO — Native-Resolution CT Foundation Model for Lung Cancer Screening (Dissertation)

- **What** — End-to-end foundation-model pipeline that classifies lung-cancer risk from screening CT at native resolution.
- **How** — Self-supervised RAD-DINO embeddings over 50M+ CT slices; patient-level aggregation; prevalence-matched threshold-aware decisioning; full DICOM/PACS curation on A100/Slurm HPC.
- **Impact** — Established native-resolution 2D embeddings as a compute-efficient alternative to 3D volumetric training and reduced false-positive follow-ups in experiments; first-author publication (J. Imaging Informatics in Medicine, 2025).

Virtual-Eyes — Quantitative CT Quality-Control Pipeline for Foundation-Model Risk Prediction

- **What** — Automated QC layer that screens CT quality before foundation-model cancer-risk prediction.
- **How** — Quantitative, rule-driven QC metrics validated against downstream model behavior on curated screening data.
- **Impact** — Quantitatively validated gains in the reliability of foundation-model cancer-risk prediction; accepted to MIDL 2026 (first author).

Multimodal Pathology Retrieval & Safer Auto-Captioning (Mayo Clinic)

- **What** — Content-based image retrieval and automated captioning for histopathology whole-slide images.

- **How** — Vision foundation-model embeddings + hybrid rule-based + LLM summarization; retrieval-guided generation to constrain and ground outputs.
- **Impact** — >0.82 cosine-similarity semantic alignment and fewer unsafe/hallucinated captions; Top-Ten Poster (Mayo AI Summit 2025); accepted to Ital-IA 2026(first author).

Federated LDCT Screening Across Five Institutions (Site Lead)

- **What** — Privacy-preserving, multi-site model training for low-dose CT lung-cancer screening.
- **How** — NVIDIA FLARE; secure model updates; leakage-safe cross-site splits; harmonized multi-institution pipelines.
- **Impact** — Improved detection performance and reduced projected clinical workload without sharing raw PHI; Champion in 3 of 5 categories, CMU–NVIDIA Hackathon 2026.

Synthetic-Phantom Longitudinal Validation of Commercial AI (Riverain VS-CT)

- **What** — Framework to measure reliability, measurement reproducibility, and drift/stability of a deployed commercial nodule-detection AI over time.
- **How** — MATLAB phantom modeling injecting synthetic pulmonary nodules into real DICOM CT volumes; purpose-built database + analysis workflow; IRB-approved prospective-validation protocol.
- **Impact** — Delivers a reproducible, auditable continuous-evaluation methodology for clinical-AI monitoring — directly aligned with GMLP and post-deployment surveillance.

Crop-Guided Diffusion for Histopathology Tissue Synthesis (NeurIPS 2025, co-author)

- **What** — Generative model synthesizing heterogeneous tissue to augment histopathology datasets.
- **How** — Semantic and visual crop-guided diffusion.
- **Impact** — Peer-reviewed acceptance at NeurIPS 2025, advancing data augmentation for pathology AI.

Systematic Review — Deep Learning for Low-Dose CT Lung Cancer Screening

- **What** — Comprehensive review of DL for LDCT screening across input dimensionality, workflow integration, and foundation models.
- **Impact** — Preprint (2026) synthesizing the state of the art and open research gaps for the field.

PUBLICATIONS & SCHOLARLY WORK

Peer-Reviewed Publications

1. Hoq ME, Tarbox L, Johann D Jr, Larson-Prior L, Prior F. *Harnessing Native-Resolution 2D Embeddings for Lung Cancer Classification: A Feasibility Study with the RAD-DINO Self-Supervised Foundation Model*. Journal of Imaging Informatics in Medicine, 2025.
2. Alfasy S, Uegami W, Hoq ME, Alabtah G, Tizhoosh HR. *Semantic & Visual Crop-Guided Diffusion for Heterogeneous Tissue Synthesis in Histopathology*. Advances in Neural Information Processing Systems (NeurIPS), 2025.

Accepted / In Press

3. Hoq ME, Larson-Prior L, Prior F. *Virtual-Eyes: Quantitative Validation of a Lung CT Quality-Control Pipeline for Foundation-Model Cancer-Risk Prediction*. Medical Imaging with Deep Learning (MIDL), accepted 2026.
4. Hoq ME, Uegami W, Alfasy S, Alabtah G, Malakshan S, Kazemi A, Schmitgen AT, Prior F, Tizhoosh HR. *Retrieval-Guided Generation for Safer Histopathology Image Captioning*. Ital-IA, accepted 2026.

Preprints & Technical Reports

5. Hoq ME, Prior F. *Artificial Intelligence for Low-Dose CT Lung Cancer Screening: A Systematic Review of Deep Learning, Input Dimensionality, Workflow Integration, and Foundation Models*. Preprints.org, 2026. doi:10.20944/preprints202606.1208.v1.
6. Mu J, Karna AK, Kumar Koushik T, Winchell J, Yang TJ, Zhang CM, Baker J, Hagen E, Hoq ME, Kim K, ... Busby B. *Towards Federated Learning Across Biobanks: Prototype Software from the 2026 Carnegie Mellon University–NVIDIA Hackathon*. OSF Preprint, 2026.

Published Abstracts & Conference Proceedings

7. Hoq ME, Uegami W, Alfasy S, Malakshan SR, Alabtah G, Kazemi A, et al. *From CNN to Vision Foundation Models and LLMs: A Multimodal Framework for Pathology Image Retrieval and Auto-Captioning*. Mayo Clinic Proceedings: Digital Health, 3(4), 2025 (Abstract).
8. Kolb S, Sunkara H, Chavane C, Hoq ME, Ali M, Xu H, Kashyap R. *Renal Replacement Therapy Following IV Contrast Administration in Critically Ill Patients with CKD*. Critical Care Medicine, 53(1), 2025 (Abstract).
9. Barnes B, Hoq ME, Binette E, Lee S, Arumairaj A. *Impact of an Elevated Fluid-Accumulation Index in Septic Patients with HFrEF: A MIMIC-IV Analysis*. Critical Care Medicine, 2026 (Abstract).
10. Hoq ME, Prior F. *Optimizing and Harnessing 20 Million NLST CT Lung-Screening Images for Robust Foundation-Model Training*. NIH Early Detection Research Network (EDRN), 2024 (Abstract).

11. Hoq ME, Yoshida S, Jung H. *Measurement of Thickness and Preservation of Thin Films at Maximum Wrapping Speed*. APTEC Annual Meeting, 2019.

Thesis

12. Hoq ME. *An Investigation of Image Correlation for Real-Time Monitoring of Wrapping-Film Deformation*. Master's Thesis, Southeastern Louisiana University, 2020.

SELECTED INVITED TALKS & POSTERS

- UAMS Radiology Retreat, 2026 — Synthetic CT Phantom for Nodule-Detection Validation.
- UAMS Winthrop P. Rockefeller Cancer Institute, 2026 — From CT Quality Control to Structured Radiology Reporting & Validation.
- NCI Early Detection Research Network (Caltech), 2024 — Optimizing & Harnessing 20M NLST CT Lung-Screening Images.
- APTEC Annual Meeting, 2019 — Digital Image Correlation for LDPE Deformation (audience 200+).

GRANTS & LARGE-SCALE RESEARCH INITIATIVES (CONTRIBUTOR)

- **NCI U24 CA215109 — The Cancer Imaging Archive (TCIA) Sustainment & Scalability**. Contributor to large-scale imaging-data infrastructure supporting multi-institutional AI research and data sharing.
- **UAMS Team Science Award (\$100K)**. Lead and main contributor to an interdisciplinary initiative on clinical AI and imaging analytics.
- **UAMS Team Science Award (\$40K)**. Lead and main contributor to a collaborative program supporting translational imaging and data-driven clinical studies.
- **EU EuCanImage (SEP 210631720; €805,479)**. Contributor to an international imaging-AI ecosystem for multi-center data integration and cancer research.
- **MOTIE Korea International R&D Program (P0006842)**. Contributor to global technology collaboration in imaging and data-driven innovation.

HONORS & AWARDS

- **Champion (3 of 5 Categories)** — CMU–NVIDIA Federated Learning Hackathon, 2026
- **Top-Ten Poster** — Mayo Clinic AI Summit, 2025
- **Second Place** — Datathon.org (Emory University School of Medicine), 2024
- Travel Grants — SIIM (2025), SCCM (2024–2025), HITLAB (2024)
- Merit Scholarship — World University of Bangladesh

PROFESSIONAL SERVICE & LEADERSHIP

Editorial & Peer Review

- Ethics Reviewer — NeurIPS 2026
- Program Reviewer — Society for Imaging Informatics in Medicine (SIIM) Annual Conference, 2025–present
- Ad Hoc Reviewer — Frontiers in Medicine, 2024–present (6+ manuscripts); Journal of Thoracic Disease
- Poster Judge — UAMS Summer Research Program, Winthrop P. Rockefeller Cancer Institute, 2024 (17 posters)

Conference & Project Leadership

- Area Chair (Challenges Track) — Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026
- Site Lead (UAMS) — Multi-institutional Federated Learning Project for LDCT Lung Cancer Screening (Moffitt, Vanderbilt, UCLA, U. Pittsburgh)
- Workshop Co-lead — Practical AI in Healthcare: Fine-tuning & Deployment with Open-Source Tools, Mayo Clinic, 2025

TEACHING & MENTORING

- **Research Mentorship — Medical Imaging AI, UAMS (2023–present)**. Mentored 3 Ph.D. students and 1 M.D. student in medical-imaging AI, GPU/HPC workflows, DICOM/PACS handling, experimental design, and reproducible research.
- **Graduate Teaching & Research Associate — SLU (2019–2020)**. Instructed Physics 101/102 laboratories and lectures; supported curriculum delivery and training in experimental methods.
- **Instructor & Mentor — Foundation-Year Physics & Mathematics (2010–2017)**. Taught and mentored 10,000+ pre-university students; developed problem-based learning and exam-preparation frameworks.

CERTIFICATIONS

- Certified in Lung Cancer Screening Education — American College of Radiology (ACR), 2024
- National Imaging Informatics Course (NIIC) — SIIM / AAPM / RSNA
- 20+ Professional Certifications — NVIDIA, Google, Coursera (AI, Deep Learning, Cloud Computing, Data Science)

PROFESSIONAL MEMBERSHIPS

- The MICCAI Society · Society for Imaging Informatics in Medicine (SIIM) · Radiological Society of North America (RSNA) · American College of Radiology (ACR) · Society of Critical Care Medicine (SCCM)

REFERENCES

- **Prof. Dr. Fred W. Prior** — Distinguished Professor & Chair, DBMI, UAMS · FWPrior@uams.edu · +1 (314) 303-2485
- **Dr. Lawrence Tarbox** — Associate Professor & Head of HPC, DBMI, UAMS · Lrtarbox@uams.edu · +1 (501) 603-1766
- **Dr. Donald J. Johann Jr.** — Professor (M.D.), Cancer Therapeutics, UAMS · djjohann@uams.edu · +1 (501) 526-6990